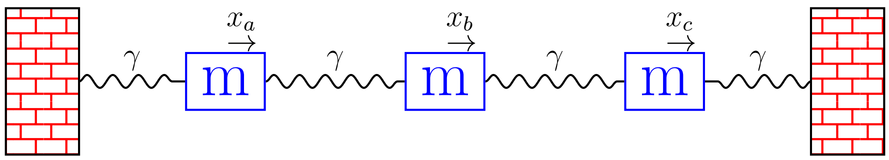
Homework 2

Coupled Oscillators

1. Conider the following Hamiltonian:

$$H = \frac{p_a^2}{2m} + \frac{p_b^2}{2m} + \frac{p_c^2}{2m} + \frac{1}{2}\gamma(x_a^2 + x_c^2) + \frac{1}{2}\gamma(x_b - x_a)^2 + \frac{1}{2}\gamma(x_c - x_b)^2$$



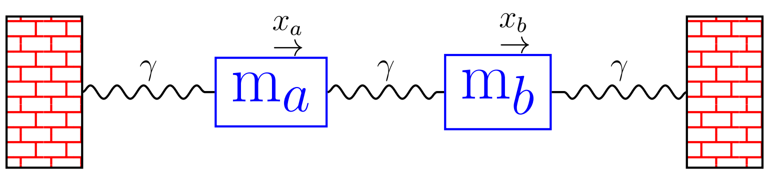
This is similar to what we did in class but we have three masses.

   (a) Construct the stiffness matrix.

   (b) Find the Eigenvalues.

   (c) Find the Eigenvectors.

2. Consider the two mass problem where the masses are different.

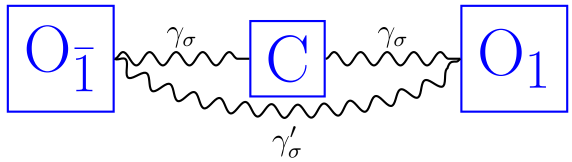


   (a) Assume that $m_a = 10m_b$; compute the Eigenvalues and Eigenvectors. Also, make a schematic of the two Eigenvectors, showing how the two masses displace in each case.

   (b) Assume that $m_a \to \infty$; compute the Eigenvalues and Eigenvectors. Also, make a schematic of the two Eigenvectors, showing how the two masses displace in each case.

3. In class we introduced the Variational Theorem. For the two-mass problem, we showed that the each Eigenvector only has a single degree of due to the normalization condition. For the two mass problem, plot the expectation value of the stiffness matrix $\langle\Psi|\hat{\mathbf{V}}|\Psi\rangle$ for a trial state vector $|\Psi\rangle = \cos\theta|X_a\rangle + \sin\theta|X_b\rangle$, and find the value of θ for which you find the minimum. Is this consistent with the Eigenvalue problem?

4. Consider CO_2 molecule (carbon dioxide), schematically pictured below, which is a linear molecule.



To keep matters simple, let us only allow motion in one dimension along the bond direction (i.e. like our three mass problem above). This will allow for the two stretching modes, but not the bending modes. Using group theory, one can show that there are 2 unique spring constants for 1d motion: $\gamma_\sigma, \gamma'_\sigma$. The corresponding potential is:

$$V = \frac{1}{2}\left[\gamma_\sigma(u_{O_1,x} - u_{C,x})^2 + \gamma_\sigma(u_{O_{\bar{1}},x} - u_{C,x})^2 + \gamma'_\sigma(u_{O_1,x} - u_{O_{\bar{1}},x})^2\right]$$

The experimental values of frequency are measured to be $\hbar\omega_A = 0.184eV$ and $\hbar\omega_A = 0.318eV$ (look up values of $\hbar$ to solve for ω). Please solve for the $\gamma_\sigma, \gamma'_\sigma$ which result in these two frequencies (look up values for atomic masses). Please use units of $eV/\mathring{A}^2$ when reporting these spring constants.